

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture – 06

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Recap of Previous Lecture



doubt?

Topic

Priority based Scheduling

Topic

Round Robin Scheduling



Topics to be Covered



Topic

Questions on Scheduling



#Q. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

GATE-2013

- A** This algorithm is equivalent to the first-come-first-serve algorithm
- B** ✓ This algorithm is equivalent to the round-robin algorithm. $(Q = T)$
- C** This algorithm is equivalent to the shortest-job-first algorithm
- D** This algorithm is equivalent to the shortest-remaining-time-first algorithm

	ΔT	ωT / Priority
P_1	0	0 3 6 9 12
P_2	0	0 3 6 9 12
P_3	0	0 3 6 9 12

$$T=3$$

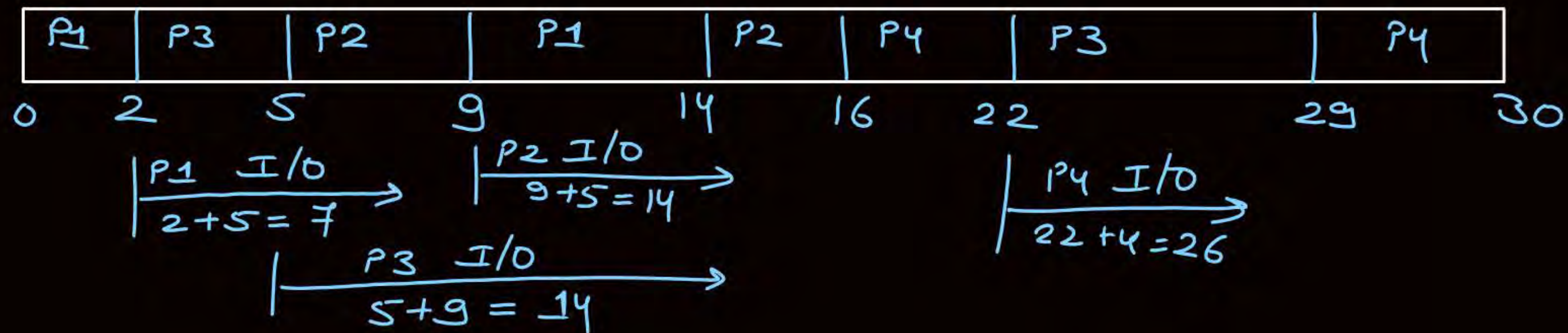
P_1	P_2	P_3	P_1	P_2	P_3	
0	3	6	9	12	15	18

- #Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses non preemptive SJF algorithm. Consider system has enough number of resources to carry out IO operations for all processes in parallel at a time. What is the average waiting time^v for the execution for the processes?

in Ready state

Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time	CT	TAT	WT
P1	0	2	5	5	14	14	2
P2	0	4	5	2	16	16	5
P3	0	3	9	7	29	29	10
P4	0	6	4	1	30	30	19

(TAT - sum of all BT)



Time	Ready Queue
0	P1 , P2, P3, P4
2	P2, P3 , P4
5	P2 , P4
9	P4, P1
14	P4, P2, P3

- #Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses preemptive SRTF algorithm. Consider system has enough number of resources to carry out IO operations for all processes in parallel at a time. What is the average waiting time for the execution for the processes?

Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time
P1	0	6	7	1
P2	1	4	2	9
P3	2	1	6	5

P1	P2	P3	P2	P1		P3	P2	P1	P2	
0	1	2	3	6	11	16	18	19	26	

$$\begin{array}{l} | \text{P3 I/O} \\ \hline | 3+6=9 \end{array} \rightarrow$$

$$\begin{array}{l} | \text{P2 I/O} \\ \hline | 6+2=8 \end{array} \rightarrow$$

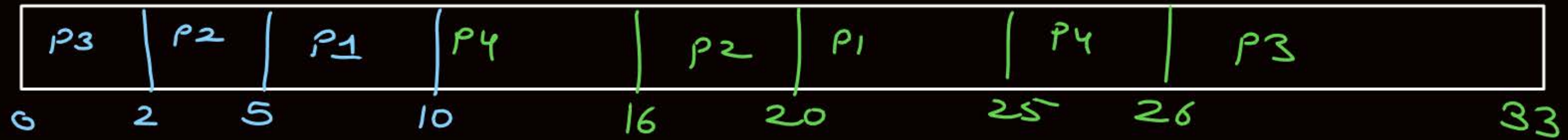
$$\begin{array}{l} | \text{P1 I/O} \\ \hline | 11+7=18 \end{array} \rightarrow$$

[MCQ]

#Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses non preemptive SJF algorithm. Consider system has only those number of resources to carry out IO operations for atmost 2 processes in parallel at a time. What is the average waiting time ^{in Ready Queue} for the execution for the processes?

R.Q.R
for I/O
overall

<u>WT</u>	Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time	CT	TAT	In R.Q. WT
11	P1	0	5	4	5(+2)	25	25	9
6	P2	0	3	7	4	20	20	6
14	P3	0	2	10	7	33	33	14
13	P4	0	6	6	1	26	26	13



$$\begin{array}{|l} 2+10=12 \\ \hline p3 \text{ I/O} \end{array} \rightarrow$$

$$\begin{array}{|l} 5+7=12 \\ \hline p2 \text{ I/O} \end{array} \rightarrow$$

$$\begin{array}{|l} 16+6=22 \\ \hline p4 \text{ I/O} \end{array} \rightarrow$$

$$\begin{array}{|l} 12+4=16 \\ \hline p1 \end{array} \rightarrow$$

12

p1 waits
for I/O
for 2 units

- #Q. The arrival time, priority and duration of the CPU and I/O bursts for each of three processes P1, P2 and P3 are given in the table below. Each process has a CPU burst followed by an I.O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	AT	Priority	BT (CPU)	BT(I/O)	BT(CPU)
P1	0	2	1	5	3
P2	2	3(lowest)	3	3	1
P3	3	1(highest)	2	3	1

The multi – programmed operating system uses preemptive priority scheduling. What are the finish times of the process P1, P2 and P3 ?

GATE - 2006

A 11, 15, 9

B 10, 15, 9

C 11, 16, 10

D 12, 17, 11

#Q. Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst Time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____.

Gate - 2019



2 mins Summary



Topic

Questions on Scheduling





Happy Learning

THANK - YOU

